

Cambridge IGCSE™

CANDIDATE
NAME

CENTRE
NUMBER

--	--	--	--	--

CANDIDATE
NUMBER

--	--	--	--



MATHEMATICS

0580/23

Paper 2 (Extended)

October/November 2021

1 hour 30 minutes

You must answer on the question paper.

You will need: Geometrical instruments

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You should use a calculator where appropriate.
- You may use tracing paper.
- You must show all necessary working clearly.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- For π , use either your calculator value or 3.142.

INFORMATION

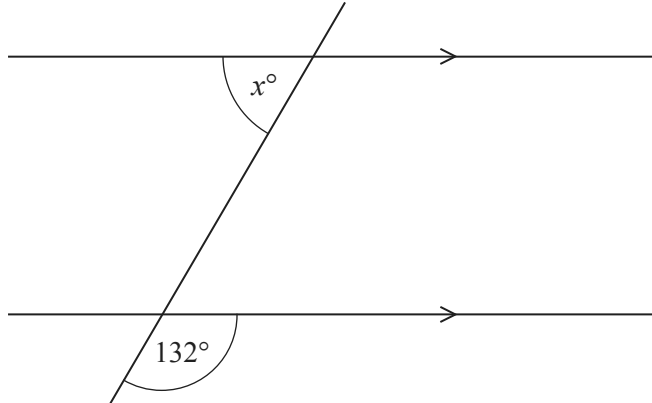
- The total mark for this paper is 70.
- The number of marks for each question or part question is shown in brackets [].

This document has **12** pages. Any blank pages are indicated.

- 1 Write 26 g as a percentage of 208 g.

..... % [1]

2



NOT TO
SCALE

The diagram shows two parallel lines intersecting a straight line.

Find the value of x .

$x =$ [2]

3

11 13 15 17 19

From this list, write down the number that is both a prime number and a factor of 195.

..... [1]

- 4 (a) = $\neq$ > <

Put a ring around each of the symbols that make this statement correct.

0.5 5% [1]

- (b) Insert one pair of brackets to make this statement correct.

$7 - 3 - 1 + 2 = 7$ [1]

- 5 Nina changes 153 euros into dollars when the exchange rate is \$1 = 0.9 euros.

Calculate the amount Nina receives.

\$ [1]

- 6 Marek buys a computer for \$420.
He sells it at a loss of 15%.

Calculate the selling price of this computer.

\$ [2]

- 7 Simplify.

$$32g^{32} \div 4g^4$$

..... [2]

- 8 Beatrice walks 1 km at a speed of 4 km/h and then 2 km at a speed of 4.5 km/h.

Work out Beatrice's average speed for the whole journey.

..... km/h [3]

- 9 Write the recurring decimal $0.\dot{2}\dot{7}$ as a fraction.

..... [1]

- 10 These are the first four terms of a sequence.

3 -1 -5 -9

- (a) Find the next term in this sequence.

..... [1]

- (b) Find the n th term.

..... [2]

11 $P = M(g^2 + h^2)$

- (a) Find the value of P when $M = 100$, $g = 3$ and $h = 4.5$.

$P =$ [2]

- (b) Rearrange the formula to write g in terms of P , M and h .

$g =$ [3]

- 12 **Without using a calculator**, work out $\frac{11}{12} + \frac{3}{4}$.

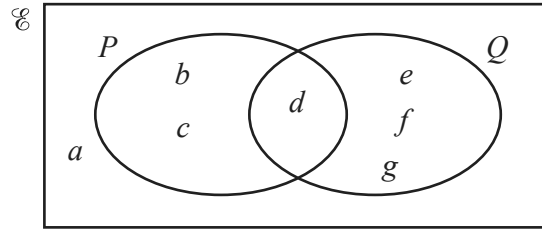
You must show all your working and give your answer as a mixed number in its simplest form.

..... [3]

- 13 Calculate $0.04^2 + 0.03 \times 0.28$.
Give your answer in standard form.

..... [2]

14



(a) Complete the statement.

$$P \cup Q = \{ \dots \} \quad [1]$$

(b) Find $n(Q)$.

..... [1]

(c) Find $n(P' \cap Q)$.

..... [1]

15 The cost of a train journey is increased by 6% to a new cost of \$153.70 .

Calculate the original cost of the train journey.

\$ [2]

16 Jo and Mo share \$26.
Jo receives \$5 more than Mo.

Find the ratio Jo's money : Mo's money.
Give your answer in its simplest form.

..... : [3]

- 17 Each interior angle of a regular polygon is 178.5° .

Calculate the number of sides of this polygon.

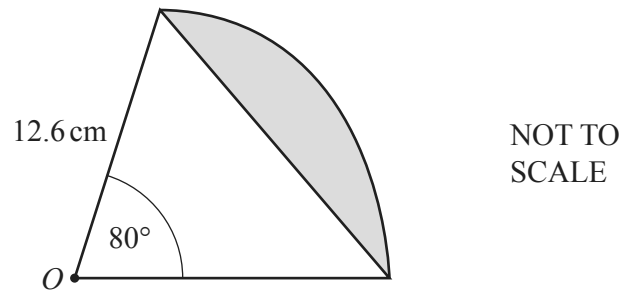
..... [2]

- 18 Find the equation of the straight line that passes through the points $(2, -2)$ and $(3, 10)$.

Give your answer in the form $y = mx + c$.

$y =$ [3]

19



The diagram shows a sector of a circle, centre O , radius 12.6 cm.

Calculate the perimeter of the shaded segment.

..... cm [4]

- 20** A lake has an area of 3 km^2 .
On a map the area of the lake is 18.75 cm^2 .

Find the scale of the map in the form $1 : n$.

1 : [3]

- 21 Simplify fully.

$$(243y^{10})^{\frac{3}{5}}$$

..... [2]

- 22 Solve the simultaneous equations.
You must show all your working.

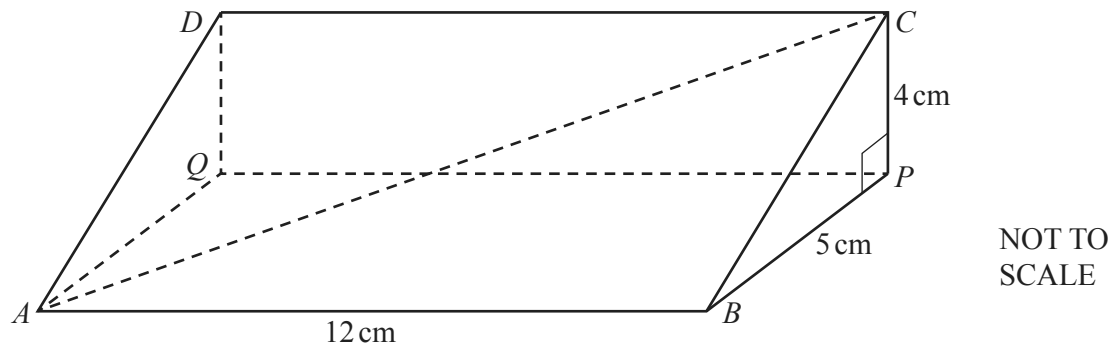
$$y = x^2 - 3x - 13$$

$$y = x - 1$$

$$x = \dots\dots\dots, y = \dots\dots\dots$$

$$x = \dots\dots\dots, y = \dots\dots\dots [5]$$

23



The diagram shows a triangular prism.
Angle $BPC = 90^\circ$.

(a) Calculate AC .

$AC = \dots\dots\dots\text{ cm}$ [3]

(b) Calculate the angle between AC and the base $ABPQ$.

$\dots\dots\dots$ [3]

24 $\tan x = \sqrt{3}$ and $0^\circ \leq x \leq 360^\circ$.

Find all the possible values of x .

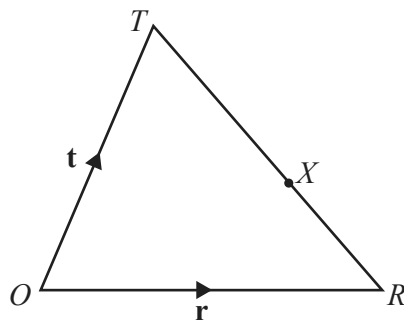
$\dots\dots\dots$ [2]

25 Simplify.

$$\frac{3x^2 - 18x}{ax - 6a + 2cx - 12c}$$

..... [4]

26



NOT TO
SCALE

ORT is a triangle.

X is a point on TR so that $TX : XR = 3 : 2$.

O is the origin, $\overrightarrow{OR} = \mathbf{r}$ and $\overrightarrow{OT} = \mathbf{t}$.

Find the position vector of X .

Give your answer in terms of $\mathbf{r}$ and $\mathbf{t}$ in its simplest form.

..... [3]

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.